

Summary of flow diagrams:

- 1) pedometer ✓
- 2) Shake feature (feed mode) ✓
- 3) IMU Data Collection ✓
- 4) UI flow ✓
- 5) hardware ✓
- 6) PCB ✓

Pedometer

read accelerometer
(a_x, a_y, a_z) /
Initialize IMU

Stand still for
3 seconds
(calibration period)

Read many IMU samples

Compute average motion
baseline

set pedometer baseline parameter

Enter main loop

play mode
active?

no

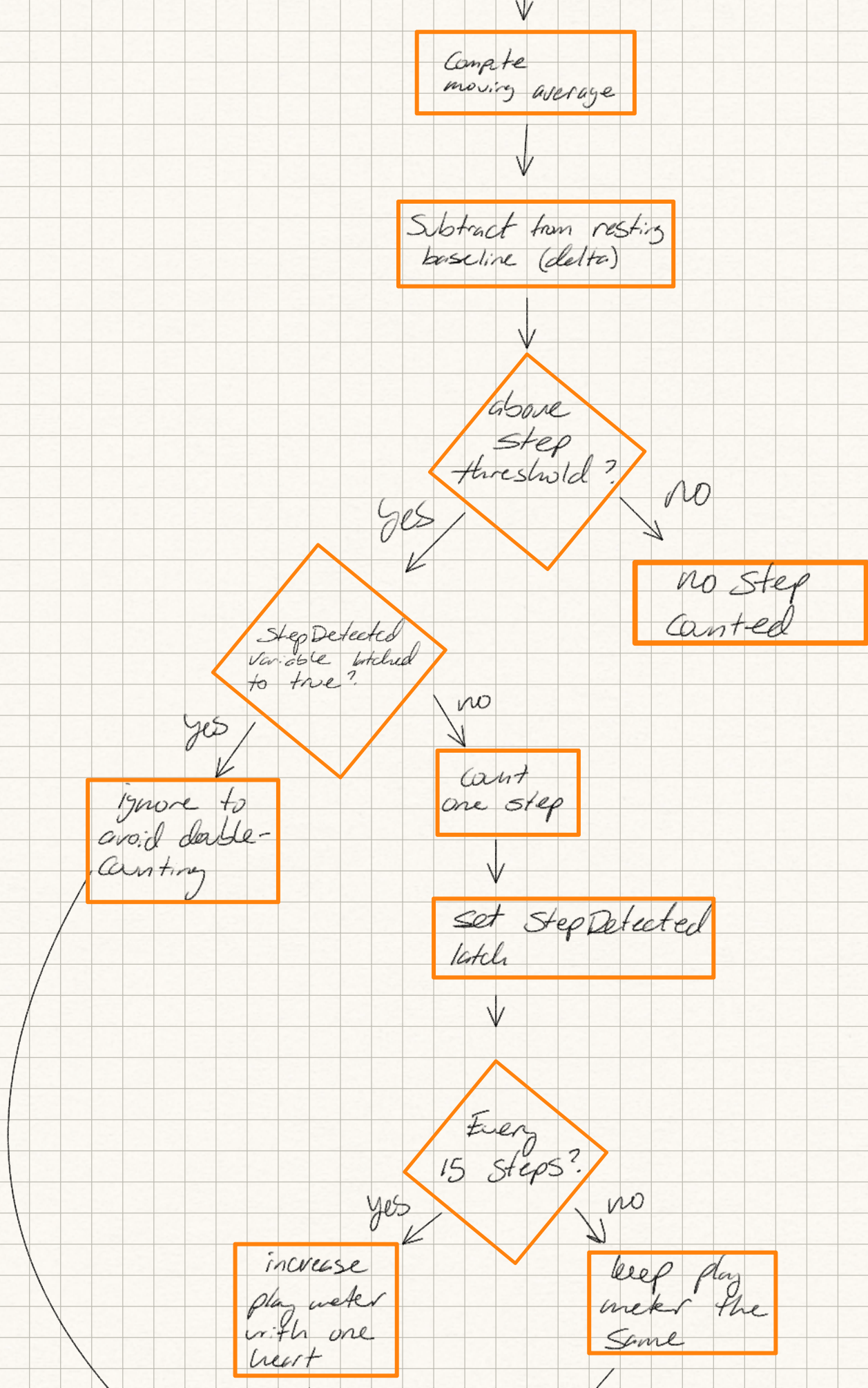
Skip pedometer
update

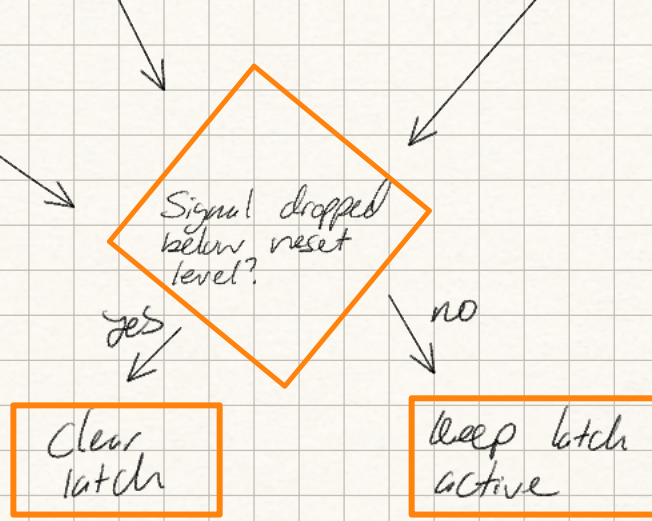
yes

read a_x, a_y, a_z
from IMU

Compute overall
motion (2D norm):
 $|a_x| + |a_y| + |a_z|$

store value in
rolling buffer (new data
overwrites oldest data once
buffer is full)





Shake feature (feed mode):

Feed mode
is active



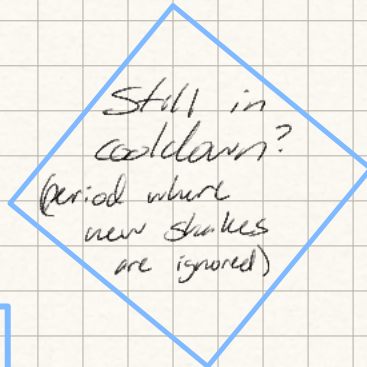
read a_y, a_x, a_z
from accelerometer (IMU)



Compare current acceleration
to previous values



Compute change in motion
magnitude

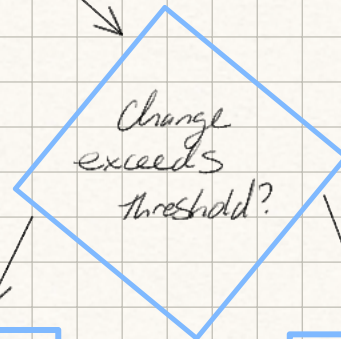


Yes



ignore shake

no



yes

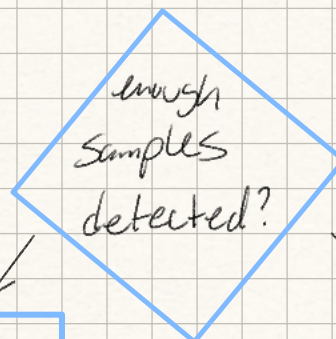


increase shake
sample counter

no



no change
in shake count



no



wait for more
motion samples

yes

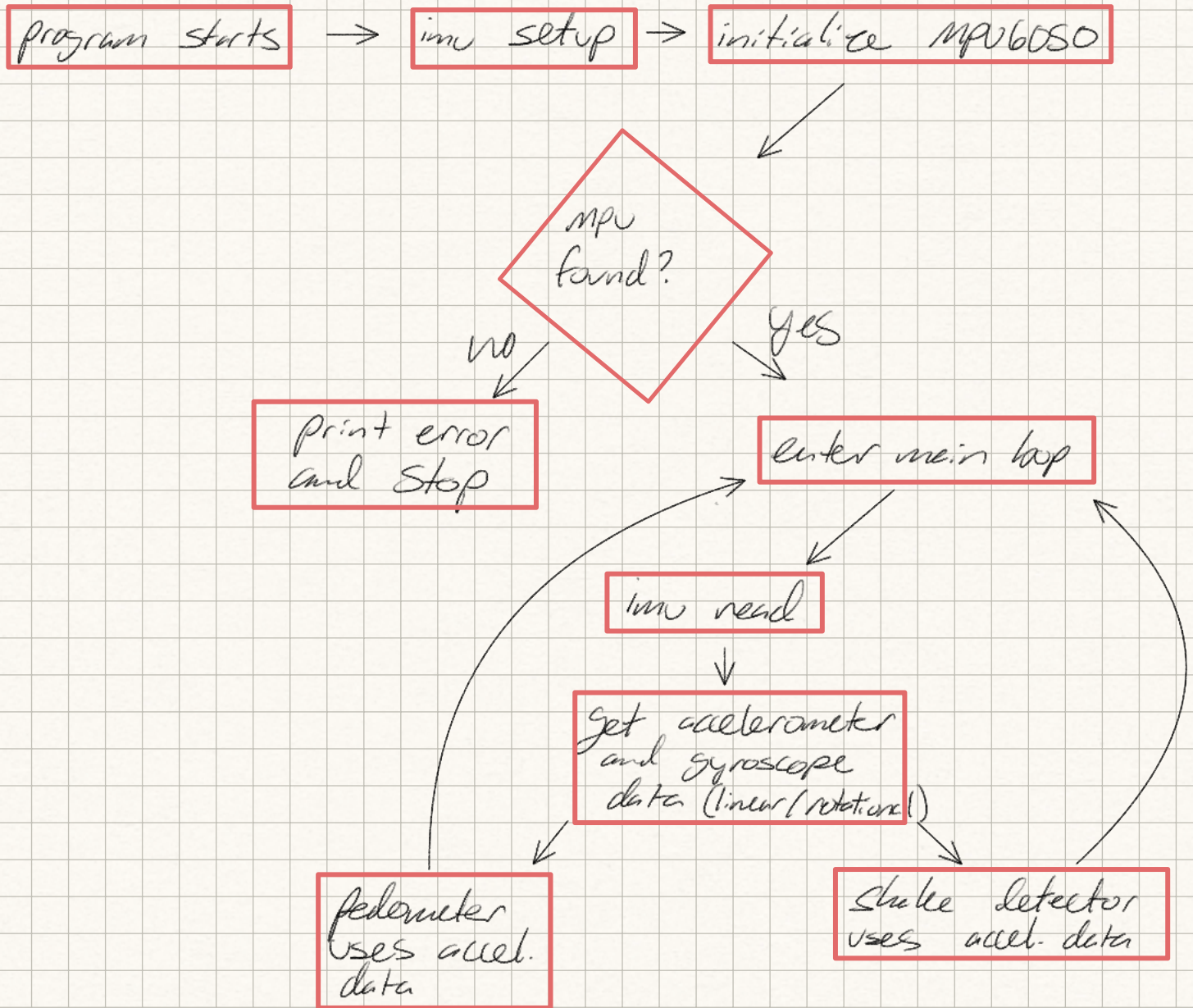


increase fullness
meter

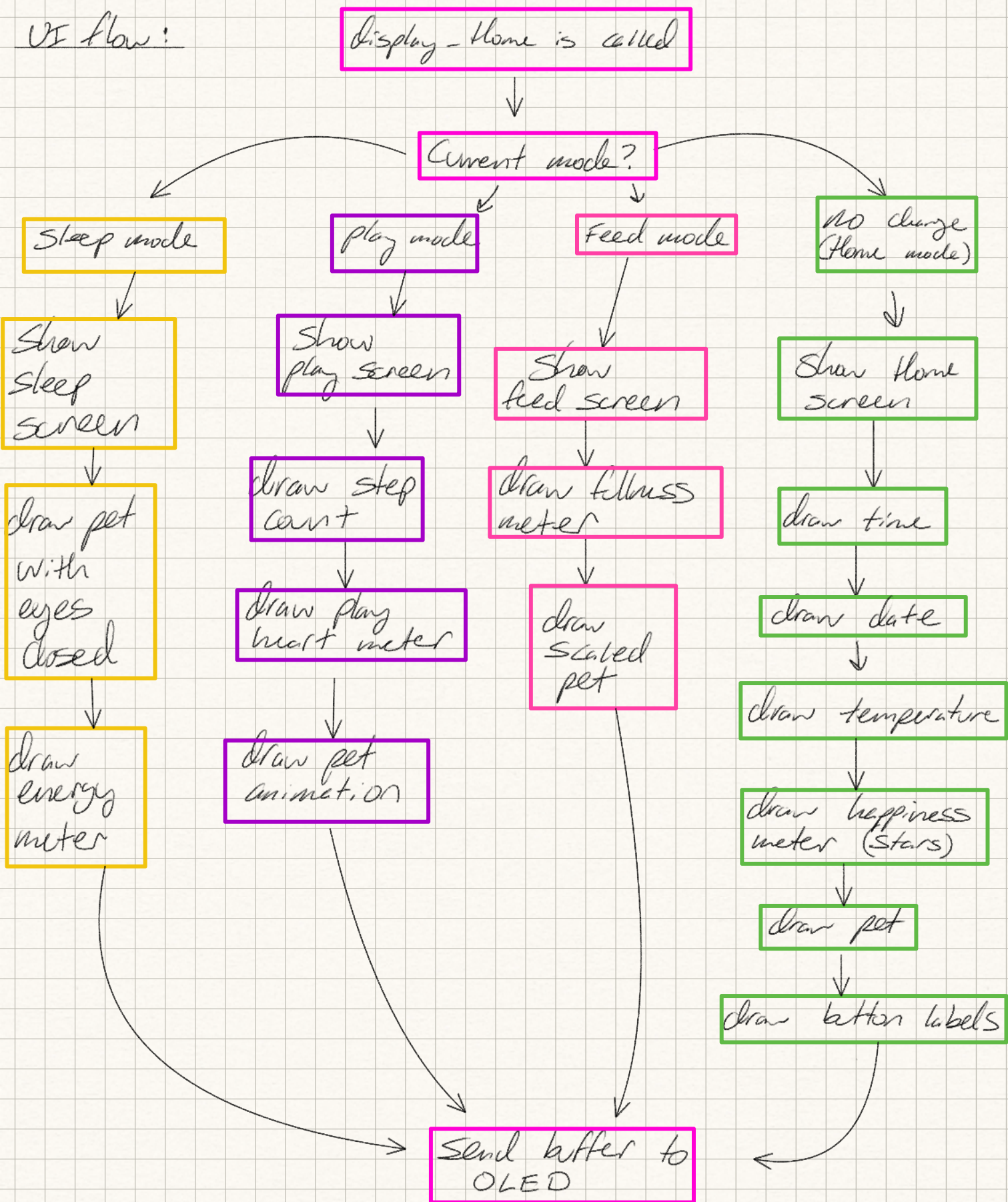


update saved state/UI

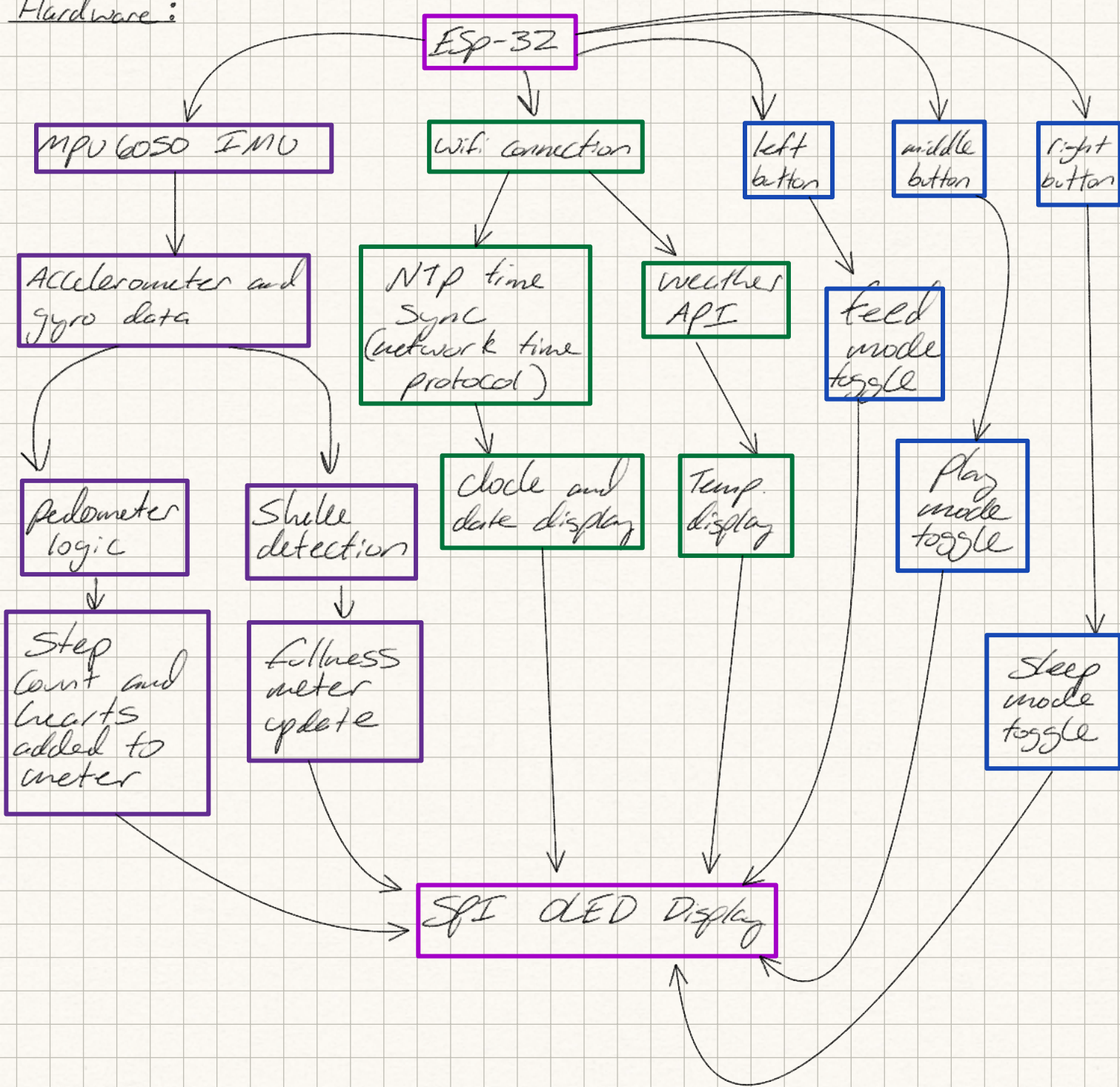
Interneal Measurement Unit (IMU) data collection:



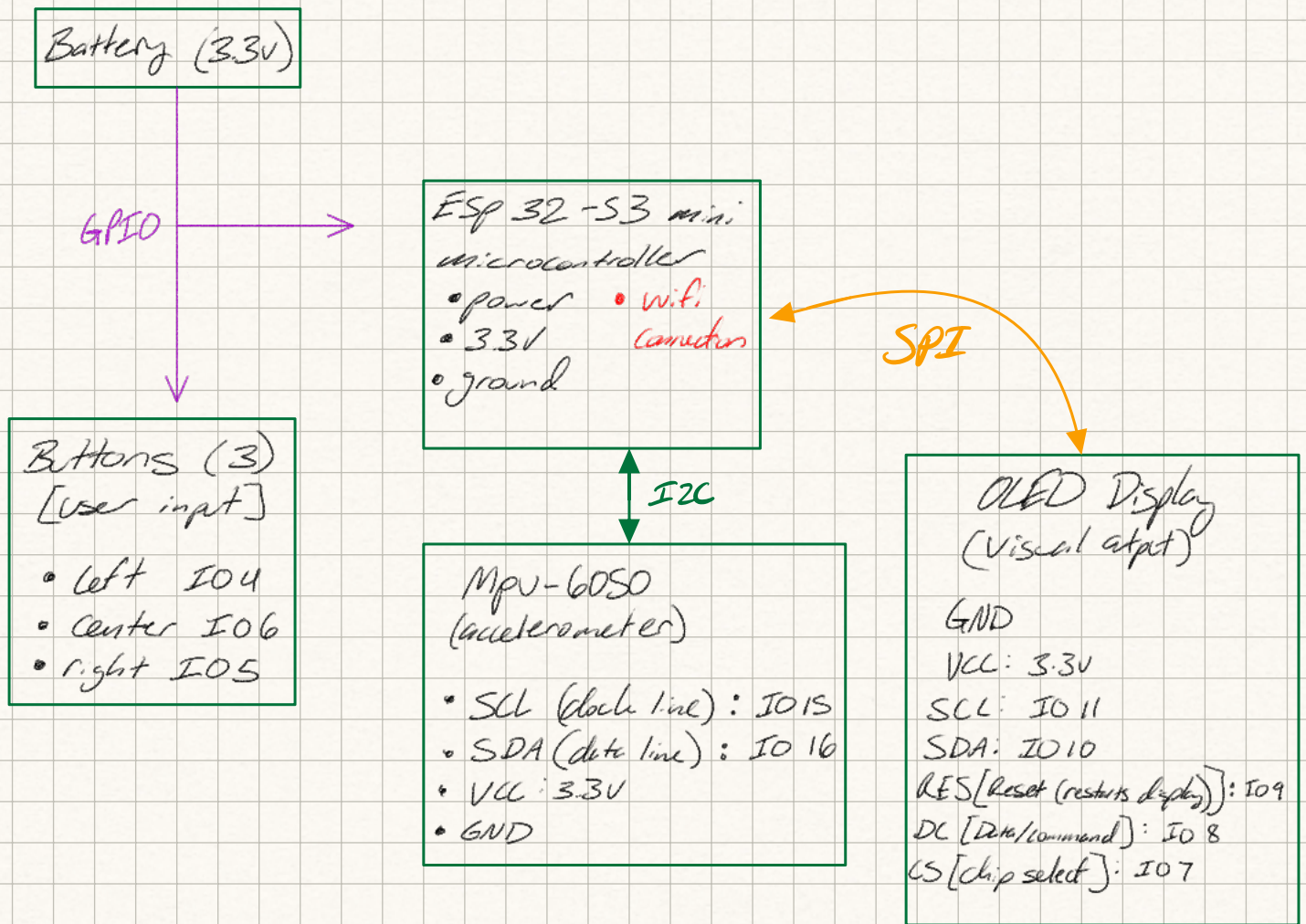
UI flow:



Hardware:



PCB:



website: non-technical

pdf document: more technical (possibly latex)

- Start with one topic
and

(link to
website)

* identify 3 things
for flow diagrams!!

- block diagram (hard
ware)

- block diagram (software)

- flow diagram (pcb)

* Send to
teacher by
tonight!!

(diagram)